روش‌های پیشرفته نمونه برداري در جنگل

نمونه برداري قطعه‌اي
Plot sampling

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مدل مفهومي نمونه برداري

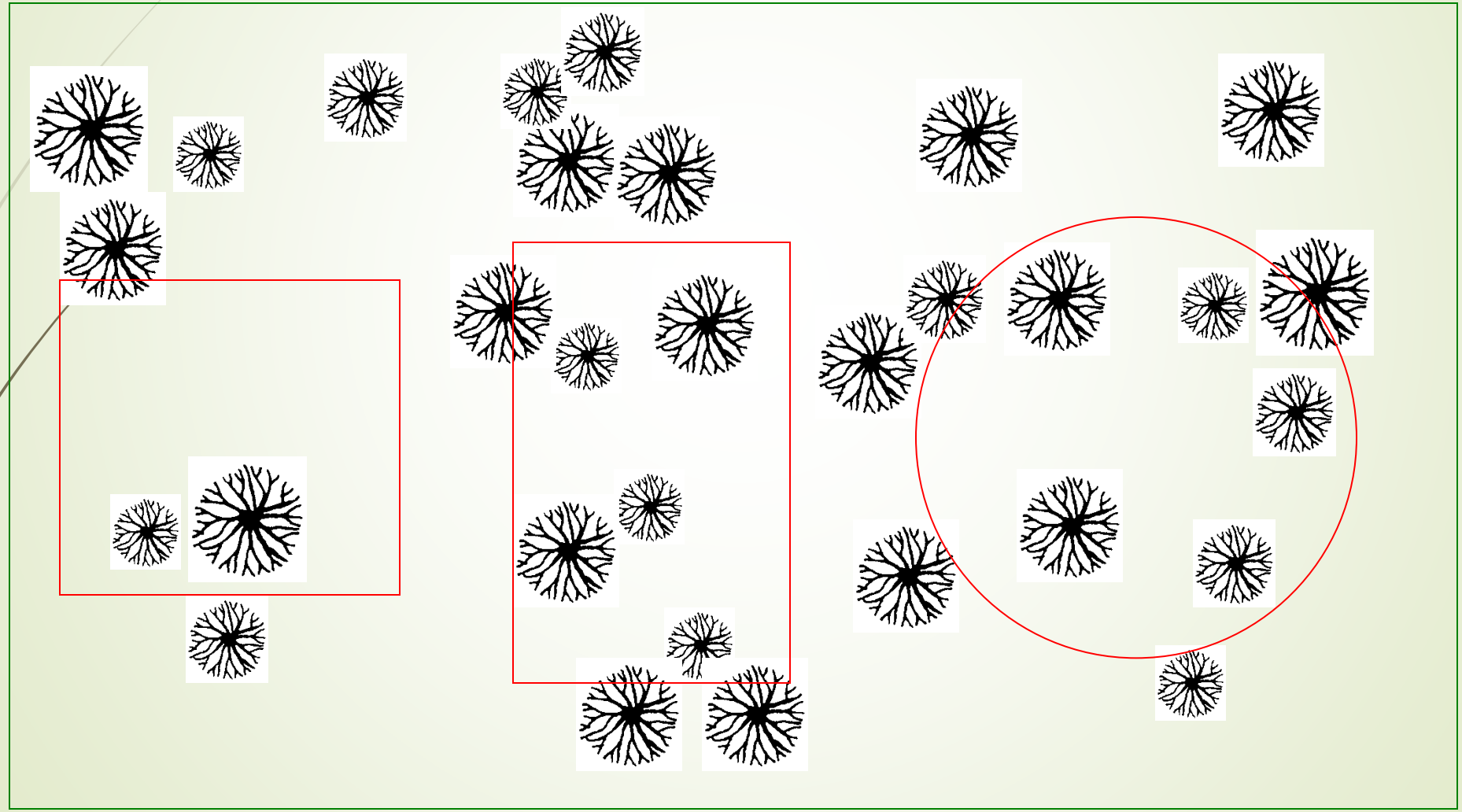
- Protocol
- طراحي نمونه برداري Sampling design
  - روش نمونه برداري Sampling Method
    - نمونه برداري نقطه‌اي Point Sampling
    - نمونه برداري فاصله‌اي Distance Sampling
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روش‌هاي رايج نمونه برداري قطعه‌اي

Plot sampling methods

- **Fixed Area Plot**
- **Multi-Scale Plot**
  - Concentric Circular Sample Plot
  - Multi-Scale Rectangular
- **Variable Area Plot**
  - Prodan (1968) Six Tree Sample plot
  - Sohrabi (2018) Adaptive K-tree Sample plot

Fixed Area Plot Sampling



Plot Shape and Size

Choose the ***optimal plot shape and size***

What do you mean by optimal:

1. Statistically
2. Ecologically
3. Logically

Plot Shape and Size

A circular plot shape has two advantages over other plot shapes:

1. It represents the geometric shape with the *smallest perimeter* for a given plot size. It tends to produce less borderline trees than plots of the same size, but different shape. (Circular < Square <Rectangular)

2. In stands without undergrowth, the *plot boundaries can be conveniently located* with the aid of optical devices, for example; with the optical Blume- Leiss or Suunto and an ultrasonic or laser (Vertex or Ledha) range-finder

Plot Shape and Size

For a given sampling intensity, the required number of plots decreases proportionally to plot size. In order to estimate the volume of a stand with an area of 15 hectares, a sampling intensity of 5%, and sample plots with a radius of 7, 10, and 13 m, respectively, the required number of sample plots is

R = 7m : n = 7500/154.0 ≈ 49
R = 10m : n = 7500/314.3 ≈ 24
R = 13m : n = 7500/531.2 ≈ 14

Relationship of error and plot size

Smith (1938) and O'Regan et al. (1966)

$$s_y^2 = k \cdot x^{-c}$$

Where

- x = plot size (in hectares),
- S^2y = variance of the target variable (basal area, volume, number of trees, etc.), expressed per hectare
- k, c (k, c > 0) representing the coefficients of the equation.

Sampling studies are required to express *c* as a function of age, site index, and forest composition. In Germany, the coefficient *c* varies between 0.3 and 0.7 and, in many cases, is close to 0.5. However, it can be expected that *k* and *c* are inversely related.

Plot Shape and Size

Freese (1961)

$$s_2^2(\%) = s_1^2(\%) \cdot \sqrt{\frac{PS_1}{PS_2}}$$

Jessen (1942)

the minimum travel distance between random sampling positions is proportional to

$$\sqrt{n}$$

Plot Shape and Size

O'Regan et al. (1966)

$$c_{\text{tot.}} = c_1 . \sqrt{n} \cdot \text{tract area} + c_2 n + c3\, N_{\text{tree}}$$

Where
c_1 = *walking cost per unit distance*
c_2 = *cost of establishing a plot*,
c_3 = *cost of measurement per tree*
√n.tract area = *approximated minimum travel distance between randomly distributed points*
n = sample size
N_{tree} = *average number of trees measured at a sampling point.*

Plot Shape and Size

Zeide (1980)

$$T = n\,(t_{tr.} + t_m)$$

Where
T = total time involved in the field work of a forest inventory, and necessary to meet the specified accuracy
t_m = measuring time and
t_{tr} = travel time

Plot Shape and Size

Wiegert (1962)

$$\text{Relative cost} = \frac{\text{Time to take one sample of a given size}}{\text{Minimum time to take one sample}}$$

$$\text{Relative variance} = \frac{\text{Variance}}{\text{minimum variance}}$$

Operational role :

pick the size with minimum product of reletive cost and relative variance

Plot Shape and Size

Hendrick (1956)

$$\hat{A} = \left(\frac{a}{1-a}\right)\left(\frac{C_0}{C_x}\right)$$

Where

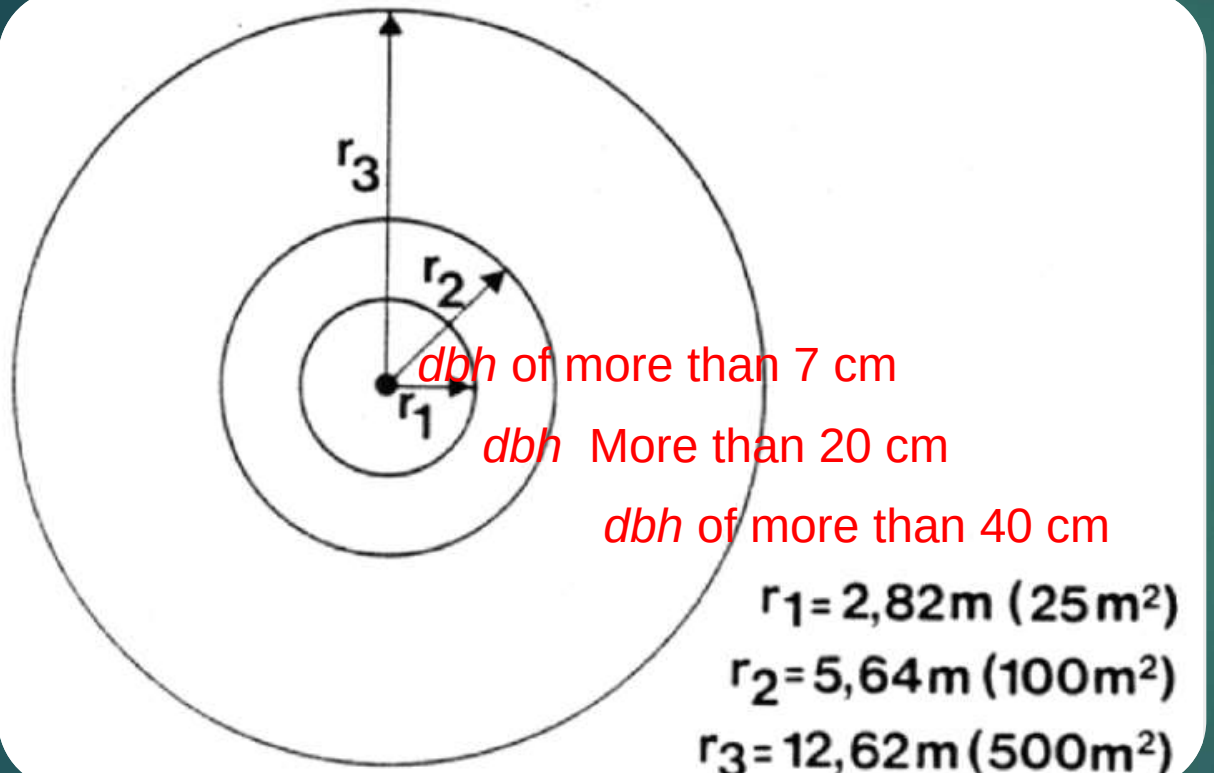
A = optimal plot size

a = absolute value of the slope of the regression on log-variance on log-PS

C_o = cost of locating one additional plot

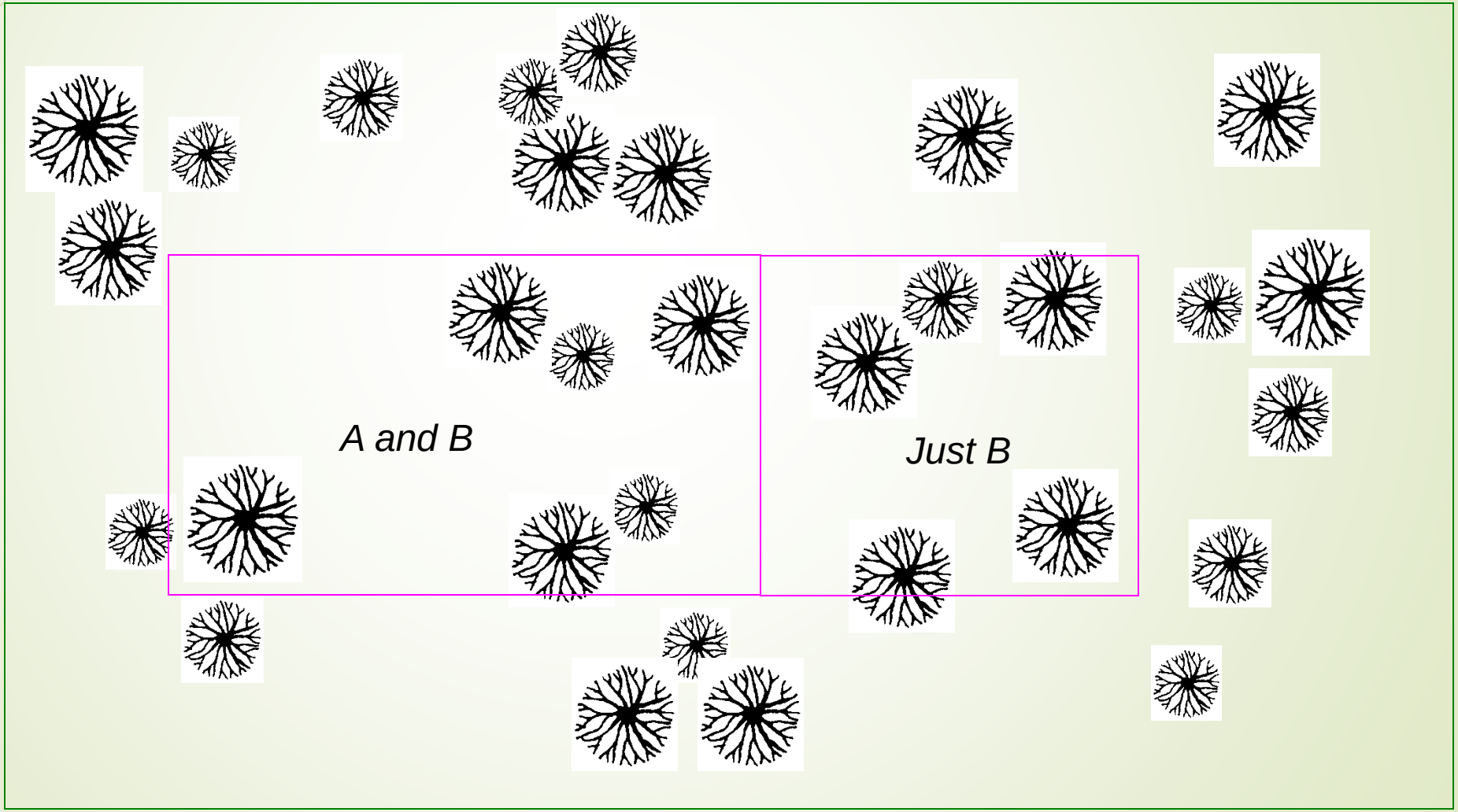
C_x = cost of measuring one unite area of sample

Concentric circular sample plot

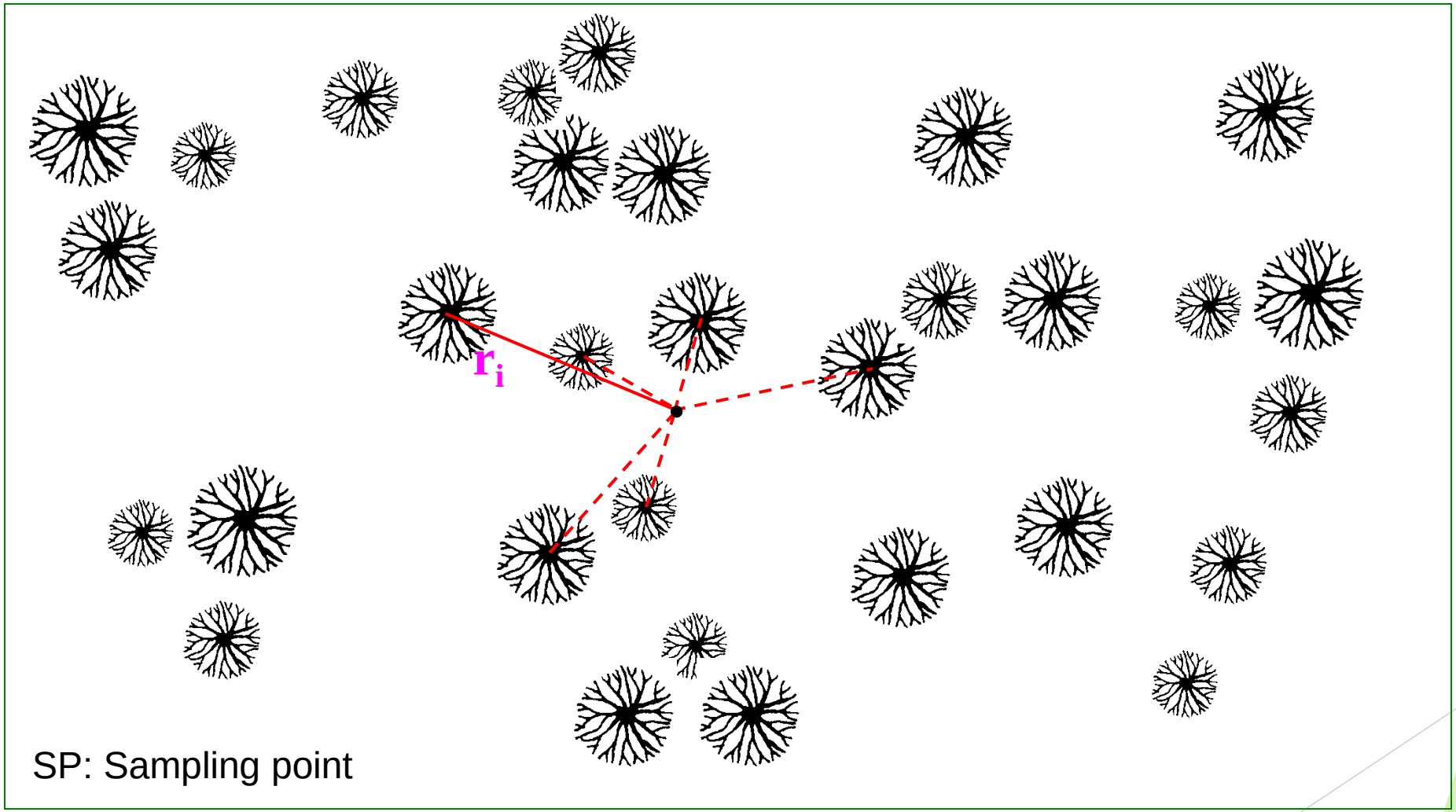


Volumes and confidence intervals have to be calculated for each of the tree size categories separately.

Multi-Scale Rectangular



Prodan (1968) Six Tree Sample plot



برآورد کننده

Prodan (1968)

$$G\,(m^2/ha) = \frac{2500}{r_6^2}\left(\sum_{i=1}^{5} d_i^2 + \frac{1}{2}d_6^2\right)$$

$$N\,(ha) = \frac{10000}{\pi r_k^2}(k - 0.5)$$

Where

d_i $(i = 1, \ldots, 5)$ = diameter (cm) of the *i*th nearest tree

r_6 =distance (m) to the *sixth* nearest tree.

Eberhardt (1967)

$$N/\text{ha} = \frac{10,000}{\pi r_k^2}(k - 1)$$

بيش برآوردي روش پرودان

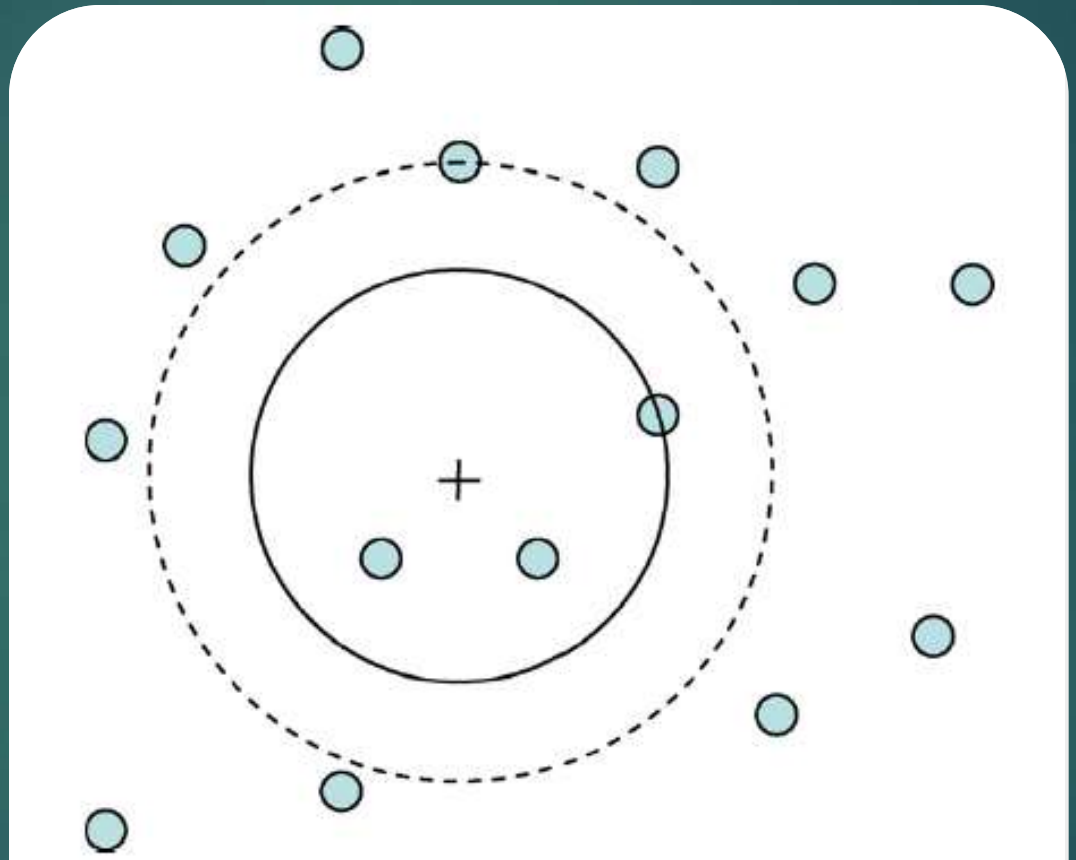


Fig. 1. Taking the radius of the distance to the k tree (here $k = 3$) as radius of the virtual sample plot (solid circle), one systematically generates the smallest possible circle plot for the k selected trees. k trees would also be contained in all circles with radii up to (but smaller than) the distance to the $(k + 1)$ tree (here: $+ 1 = 4$, dashed circle).

A new empirical approach for estimation in k-tree sampling
Kleinn and Vilcko, 2006

while the radius of the circular plot that would yield unbiased results remains unknown, we know that it should be somewhere between the distance d_k to the k tree and the distance d_{k+1} to the (k + 1) tree, because the plot radius $r = d_k$ causes a systematic overestimation and and the plot radius $r = d_{k+1}$ a systematic underestimation.

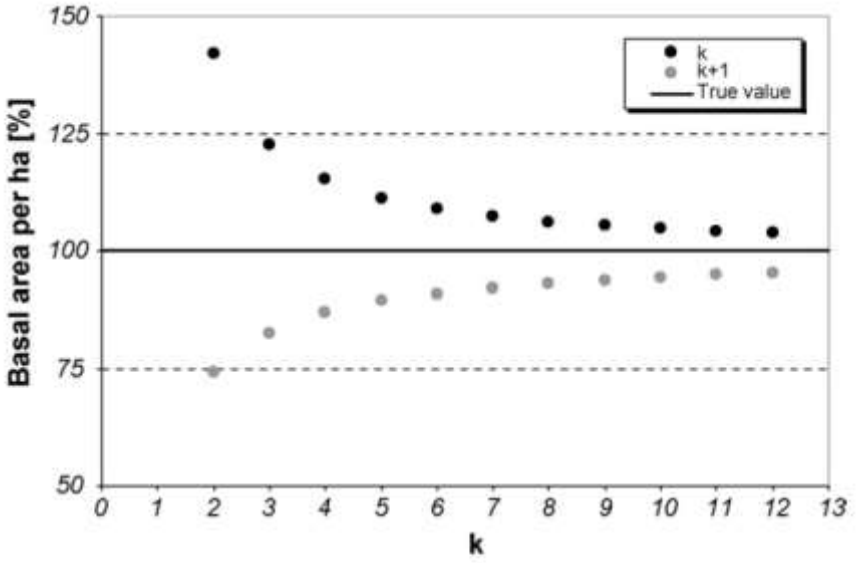


Fig. 3. Basal area estimated from *k*-tree samples in tree map (A) of Fig. 2. The virtual circular plot radius is, for each sample point, defined by the distance to the *k* tree (upper dots) and to the (*k* + 1) tree (lower dots). The true basal area value of 16.37 m^2 is here set to 100% to facilitate comparisons.

A new empirical approach for estimation in k-tree sampling
Kleinn and Vilcko, 2006

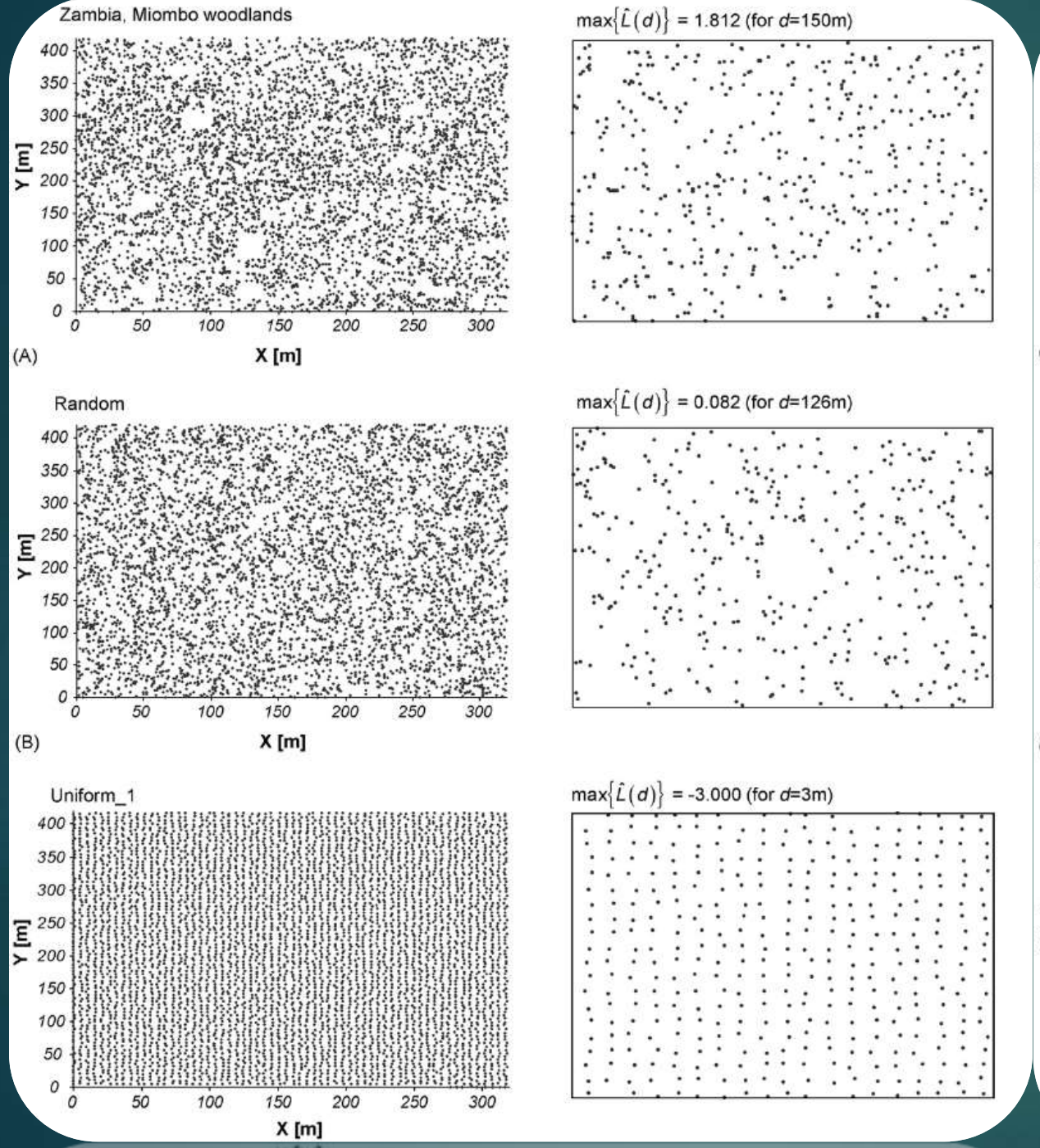
In order to find a suitable plot radius r_t on the interval $d_k < r_t < d_{k+1}$ we analyzed two simple approaches to calculate the radius of the individual plot for a fixed number k of trees:

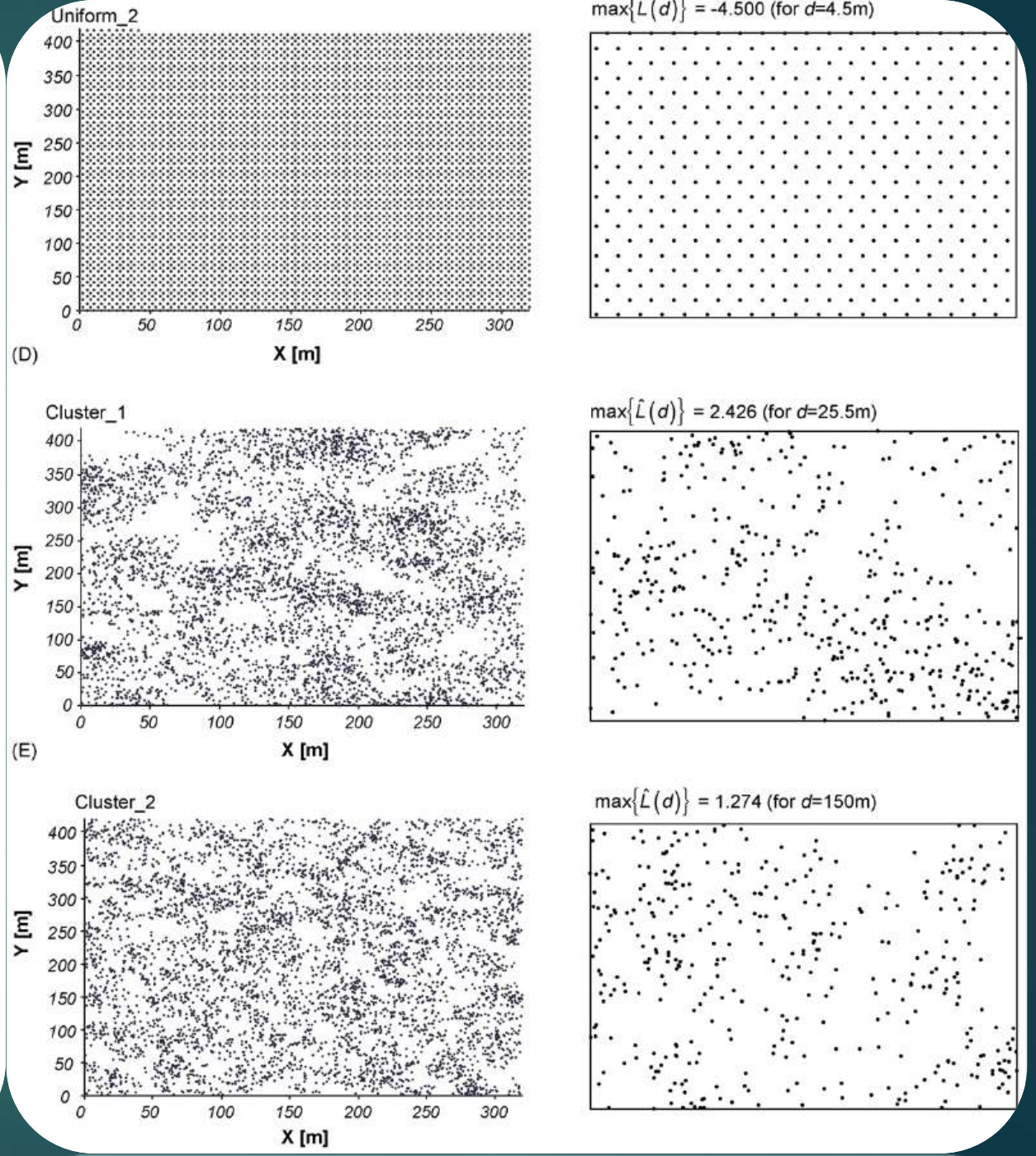
1. the arithmetic mean of the two distances to the k and to the (k + 1) tree (md = mean distance):

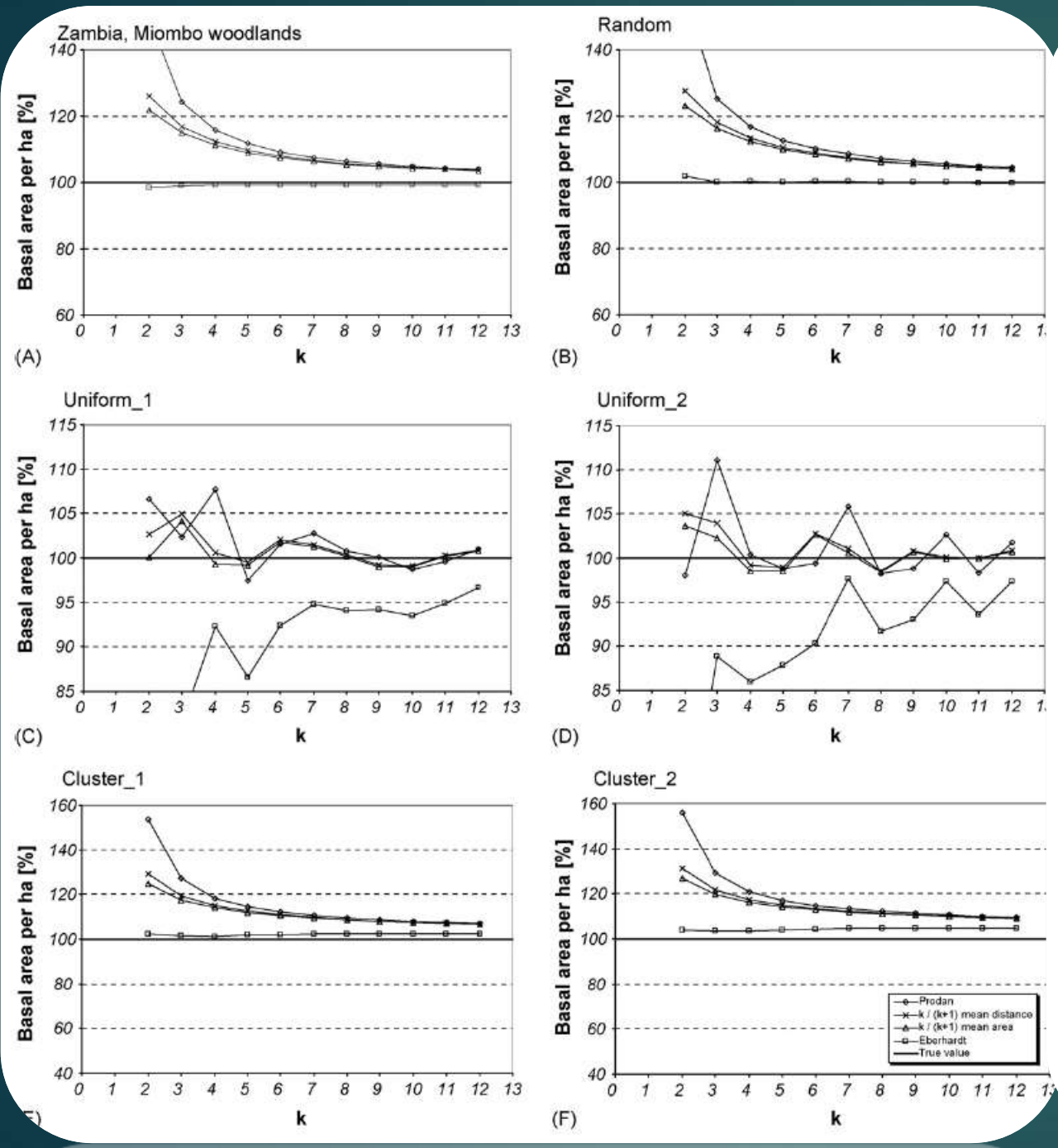
$$r_{\mathrm{md}} = \frac{1}{2}(d_k + d_{k+1})$$

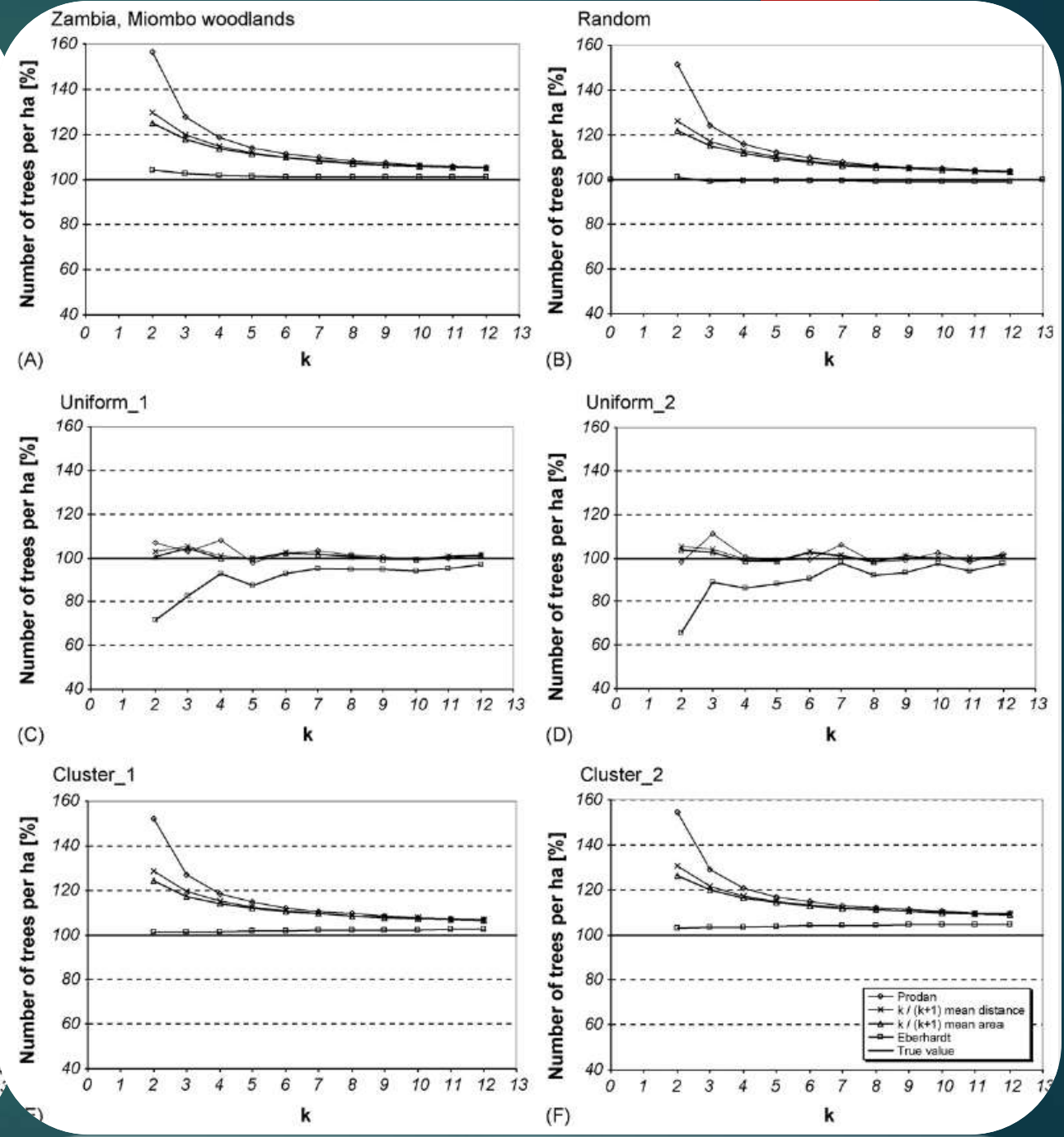
2. the radius of the average area of the circles (ma = mean area).

$$r_{\mathrm{ma}} = \sqrt{\frac{1}{2}(d^2 + d^2)}$$

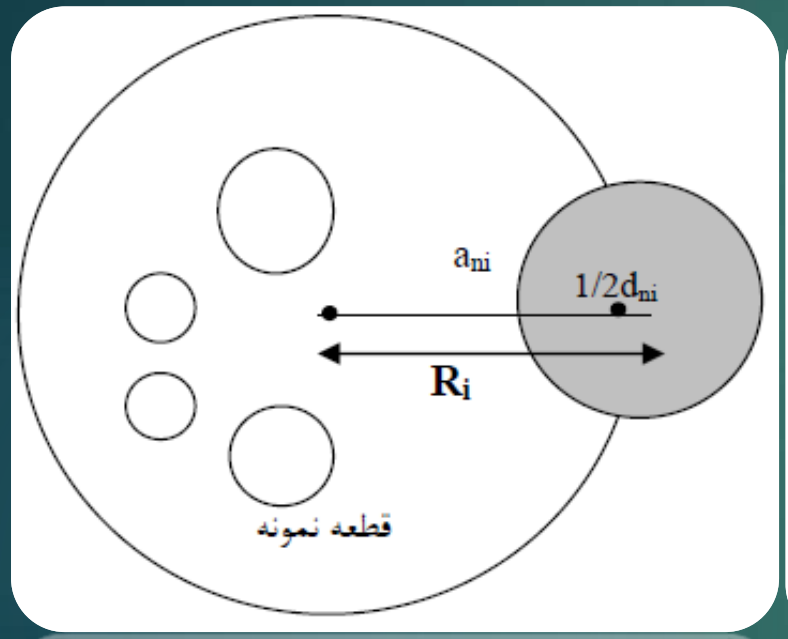

Zambia, Miombo woodlands
Y [m]
X [m]
(A)
$\max\{\hat{L}(d)\}$ = 1.812 (for d=150m)
Random
Y [m]
X [m]
(B)
$\max\{\hat{L}(d)\}$ = 0.082 (for d=126m)
Uniform_1
Y [m]
X [m]
$\max\{\hat{L}(d)\}$ = -3.000 (for d=3m)

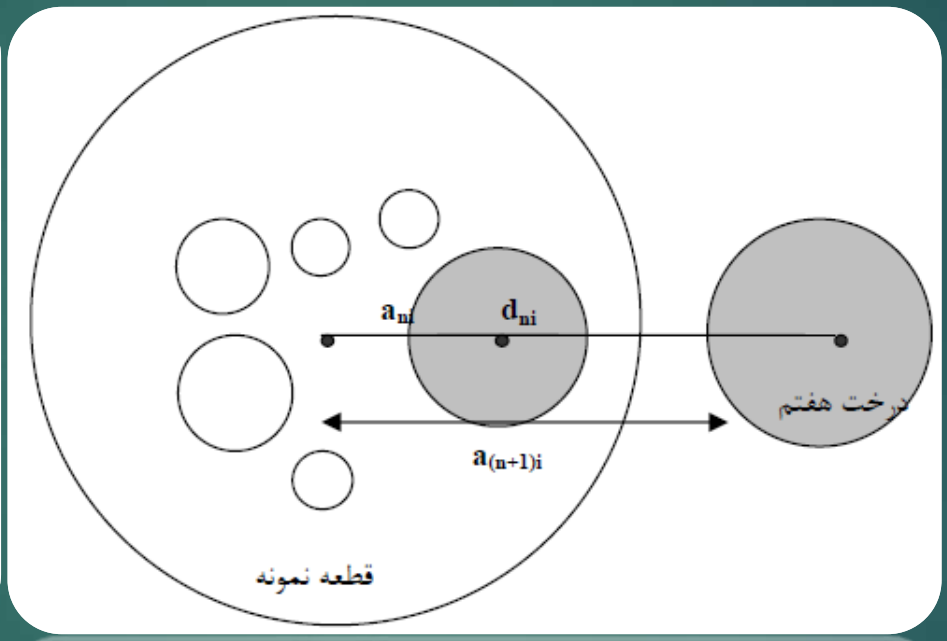

Uniform_2
Y [m]
X [m]
(D)
$\max\{\hat{L}(d)\}$ = -4.500 (for d=4.5m)
Cluster_1
Y [m]
X [m]
(E)
$\max\{\hat{L}(d)\}$ = 2.426 (for d=25.5m)
Cluster_2
Y [m]
X [m]
$\max\{\hat{L}(d)\}$ = 1.274 (for d=150m)

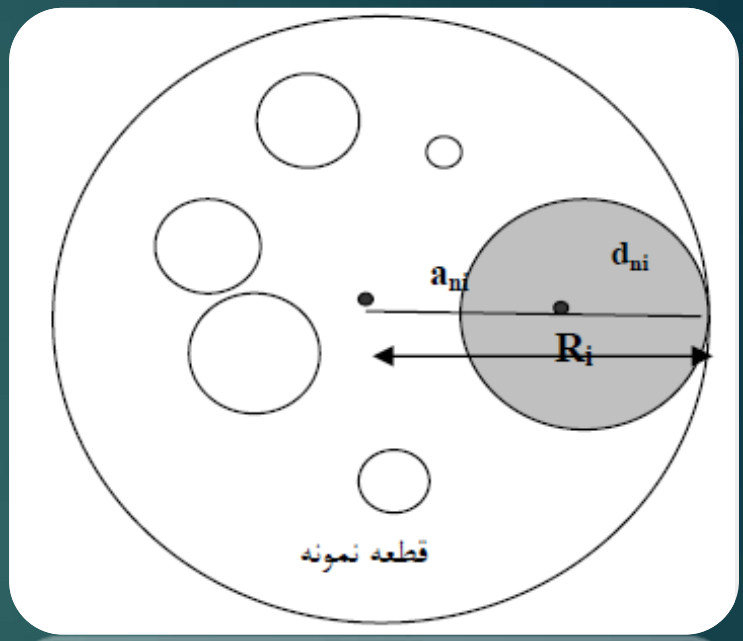

Zambia, Miombo woodlands
Basal area per ha [%]
k
(A)
Random
Basal area per ha [%]
k
(B)
Uniform_1
Basal area per ha [%]
k
(C)
Uniform_2
Basal area per ha [%]
k
(D)
Cluster_1
Basal area per ha [%]
k
Cluster_2
Basal area per ha [%]
k
(F)
Prodan
k / (k+1) mean distance
k / (k+1) mean area
Eberhardt
True value


Zambia, Miombo woodlands
Number of trees per ha [%]
k
(A)
Random
Number of trees per ha [%]
k
(B)
Uniform_1
Number of trees per ha [%]
k
(C)
Uniform_2
Number of trees per ha [%]
k
(D)
Cluster_1
Number of trees per ha [%]
k
Cluster_2
Number of trees per ha [%]
k
(F)
Prodan
k / (k+1) mean distance
k / (k+1) mean area
Eberhardt
True value

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JOURNAL OF FOREST SCIENCE, *64*, 2018 (1): 17–24

doi: 10.17221/111/2017-JFS

Adaptive *k*-tree sample plot for the estimation of stem density: An empirical approach

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(*i*) If a circular plot with the radius r_{min} is surveyed without encountering any trees, that plot is tallied as empty (Fig. 3a). r_{min} is the distance that is considered to search for reaching the first tree;

(*ii*) If at least one tree is tallied in the plot with r_{min} radius, and k trees are tallied within the r_{max} radius plot, the distance to k (or $k + 1$) tree is recorded (Figs 3c, d), r_{max} is the farthest distance that is searched to reach the kth tree.

(*iii*) If the plot with r_{max} radius is surveyed before k trees are tallied, then sampling stops and the plot is recorded as containing k_i' trees (Fig. 3b). k_i' is the number of trees on a circular plot with $r = r_{max}$ and always $k_i' < k$.

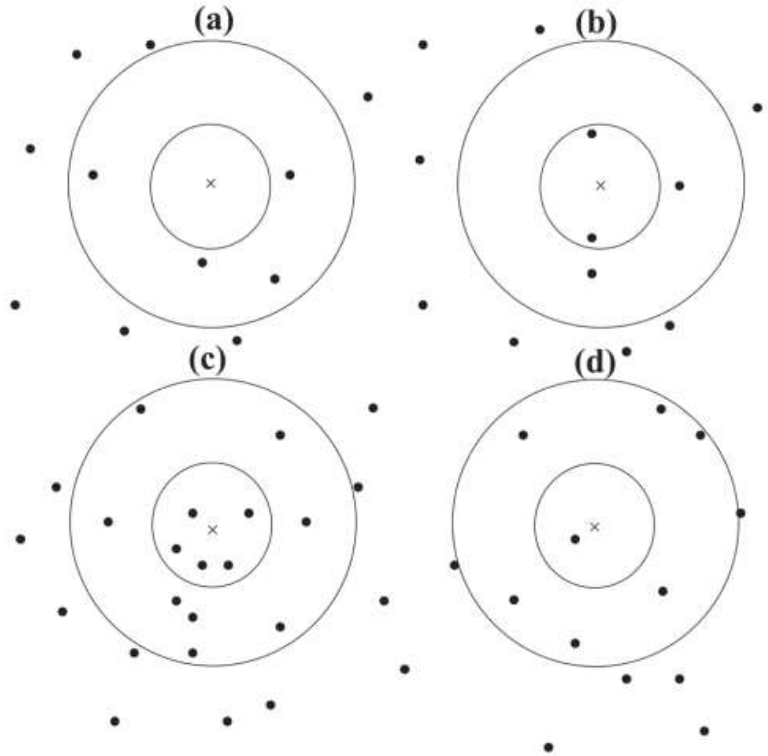


Fig. 3. Four different situations encountered for the modified adaptive approach. First, the inner circle with the radius r_{min} is surveyed. If no trees are recorded before reaching r_{min}, the plot is recorded as an empty plot with $N = 0$ and $r = r_{min}$ (a), if there is at least one tree closer than the r_{min} distance, but less than k trees are recorded in the r_{max} distance, the plot radius is r_{max} with k_i' trees (b), if k trees are tallied before reaching r_{max}, the distance to k or $k + 1$ is recorded as the plot radius (c, d)

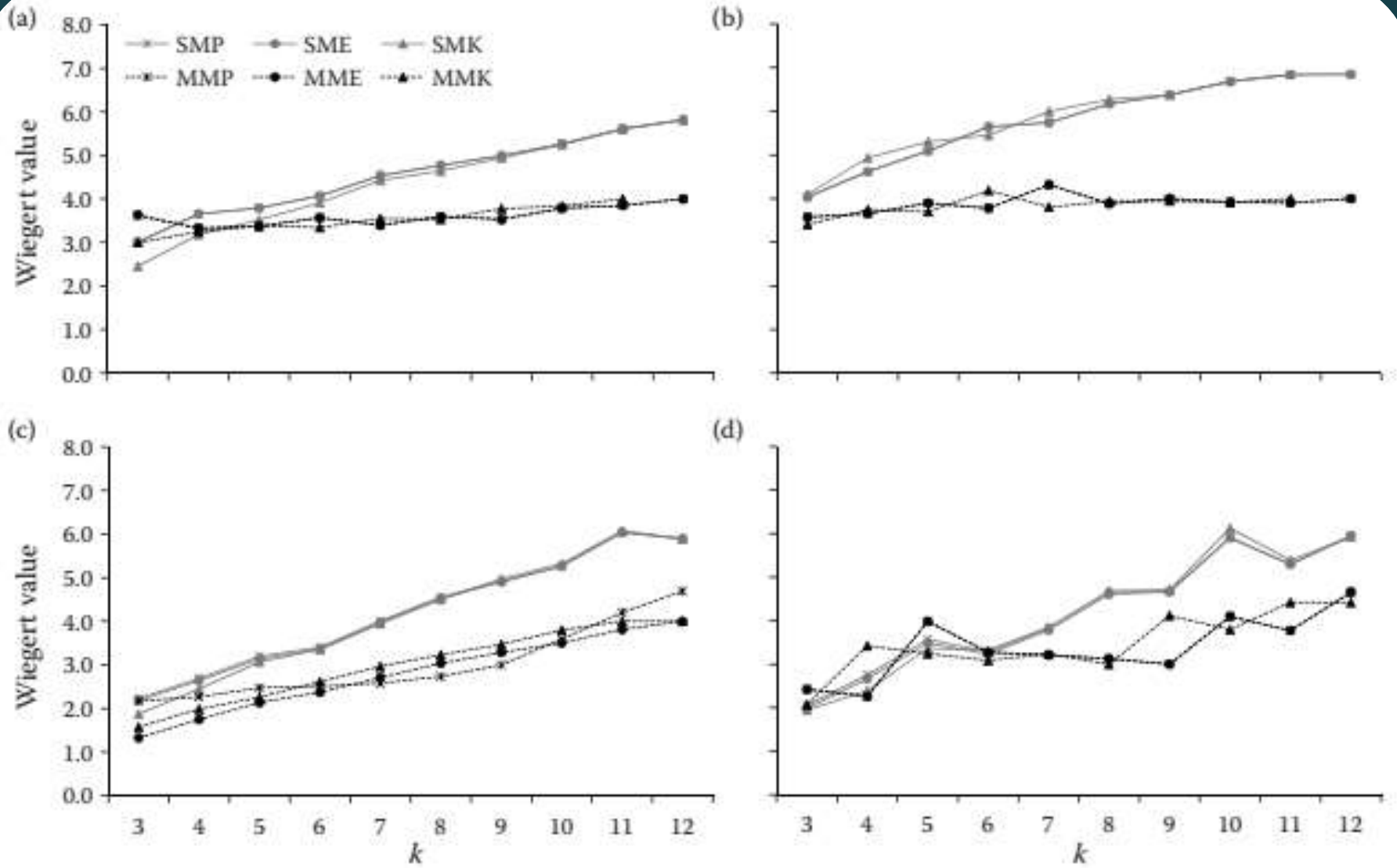


Fig. 7. Comparison of the Wiegert value of the number of efficiency for the standard and modified *k*-tree sampling for *k* = 3, ..., 12 trees using different estimators: Poisson forest (a), cluster 1 forest (b), cluster 2 forest (c), actual forest (d)

k – fixed number of trees in each sampling plot, SMP, SME, SMK – standard method with Prodan, Eberhardt and Kleinn estimator, MMP, MME, MMK – modified method with Prodan, Eberhardt and Kleinn estimator

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